

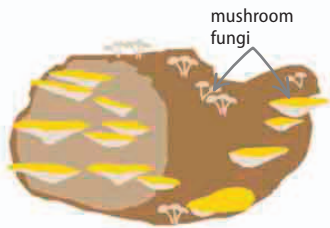
Feeding

THE PROCESS OF COLLECTING AND CONVERTING RAW MATERIALS INTO ENERGY.

Not all living things feed—plants and other photosynthetic organisms make their own food. However animals, fungi, and many single-celled organisms survive by consuming other living things.

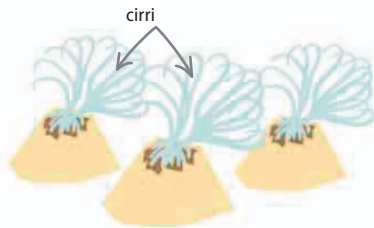
What is feeding?

An organism that feeds is called a heterotroph, a name that means “other eater.” As the name suggests, heterotrophs collect the nutrients and energy they need by consuming other organisms. Plants are called autotrophs—“self-eaters”—because they generate everything they need to survive themselves. There are several modes of feeding and every organism specializes in getting its food in a specific way.



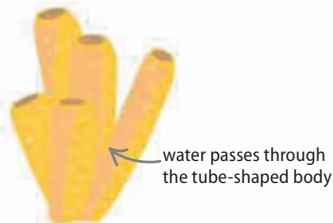
△ External digestion

A fungus is a saprophyte, meaning it grows over its food source, secreting enzymes that digest the food externally. Nutrients are then absorbed directly into its body.



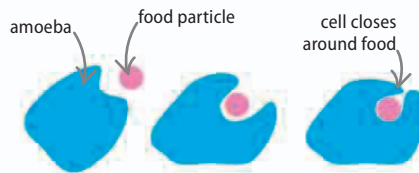
△ Filter feeding

Barnacles do not search for food, but sieve it from the water using their long, feathery legs, called cirri. Many shellfish, such as clams, are also filter feeders.



△ Absorption

The simplest feeding method is to absorb food through the surface of the body. The body of a sponge is tube-shaped and food is collected from water flowing through it.



△ Phagocytosis

Single-celled organisms such as amoebas engulf their food, moving their cell membrane around it to form a sac in which the food is digested.

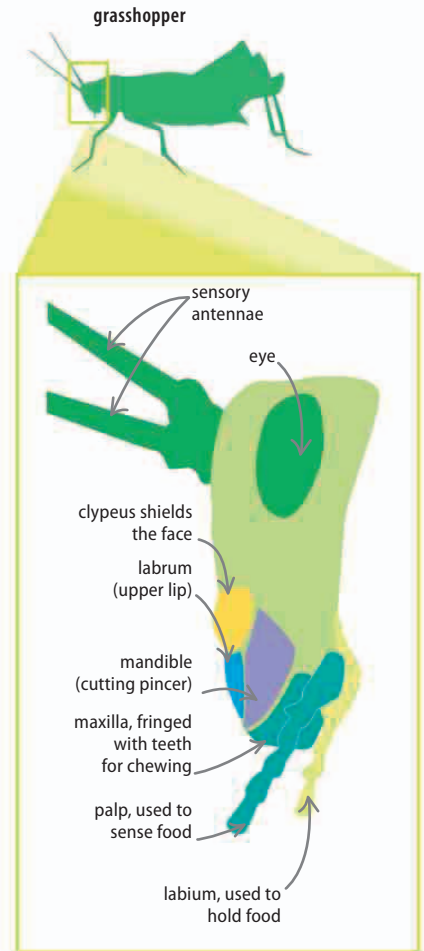


△ Biting

Only vertebrates, such as crocodiles, have jaws that open and close in a biting motion. The jaws are lined with teeth, which cut the food into manageable chunks before swallowing.

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△ Mouthparts

Insects and other arthropods have complex mouthparts. A grasshopper's mouthparts are suited to cutting and chewing, but other insects have mouthparts that can be used for sucking, biting, or soaking up liquids.